MILITARY SPECIFICATION

WIRING HARNESS: M14

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of switch assembly entitled Wiring Harness: M14, which is used on the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8808362	- Container, Packing, Ammunition, For Domestic (CONUS) Shipments of Special Weapon Items, Packing and Marking.
8813539	- Schematic Diagram, Selector, Fuze Option.
8821419	- Switch Rotary, Fuze Option M14.
8822039	- Bag, Packing, Ammunition for Selector Fuze Option.
8842517	- Gage.
EL-8842556	- Equipment List.

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PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The harness shall comply with all requirements specified on the drawing (dwg) 8821419, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

3.5 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

3.6 Switch rotation. The switch shall operate with a torque not exceeding 20 inch pounds. The action of the switch shall not permit stopping between detent positions; it shall either return to the starting position or carry through to the opposite position. These conditions shall be met when the switch is tested in accordance with 4.4.1.

3.7 Current capacity.

3.7.1 Continuous current. Each of the circuits indicated in Table I shall be capable of carrying a continuous current of 1.0 ± 0.1 ampere without interruption when tested in accordance with 4.4.2.1.

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3.7.2 Peak current. All the assemblies must be capable of meeting the requirements of 3.7.2.1 and 3.7.2.2, but any one unit need not be tested to both paragraphs after having been tested as specified in one of these paragraphs.

3.7.2.1 Clockwise. Each of the circuits indicated in Table I for a clockwise (cw) shaft position shall be capable of carrying a direct current of 50 plus 0 minus 5 amperes for 5 milliseconds without interruption with a maximum voltage drop of 2.0 volts across the circuit when tested in accordance with 4.4.2.2.1.

3.7.2.2 Counterclockwise. Each of the circuits indicated in Table I for a counterclockwise (ccw) shaft position shall be capable of carrying a direct current of 50 plus 0 minus 5 amperes for 5 milliseconds without interruption with a maximum voltage drop of 2.0 volts across the circuit, when tested according to 4.4.2.2.2.

Table I

(For circuit reference see dwg 8813539)

Switch Position	Circuit Between Connector Pin	and Pin
cw	A	B
cw	E	F
cw	H	J
ccw	B	C
ccw	F	G
ccw	J	D

3.8 Circuit resistance. The resistance of each circuit shall not exceed the maximum value listed in Table II when tested in accordance with 4.4.3.

Table II

(For circuit reference see dwg 8813539)

Switch Position	Circuit Between Connector Pin	to Pin	Maximum Resistance	Test Code No.
cw	A	B	0.04 ohms	050001
cw	E	F	0.04 ohms	050002
cw	H	J	0.04 ohms	050003
cw	K	L	0.04 ohms	050004
ccw	B	C	0.04 ohms	050005
ccw	F	G	0.04 ohms	050006
ccw	J	D	0.04 ohms	050007

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3.9 Insulation resistance. The insulation resistance of each terminal to all other terminals listed in Table III shall be greater than 20 megohms when tested at 500 + 50 volts direct current (vdc). Testing shall be as specified in 4.4.4.

Table III

Switch Position	Between Pin	and Pin	Test Code No.
CW	A	C	030001
CW	A	E	030002
CW	A	G	030003
CW	A	K	030004
CW	A	H	030005
CW	A	D	030006
CW	A	Case	030007
CW	C	E	030008
CW	C	G	030009
CW	C	K	030010
CW	C	H	030011
CW	C	D	030012
CW	C	Case	030013
CW	E	G	030014
CW	E	K	030015
CW	E	H	030016
CW	E	D	030017
CW	E	Case	030018
CW	G	K	030019
CW	G	H	030020
CW	G	D	030021
CW	G	Case	030022
CW	K	H	030023
CW	K	D	030024
CW	K	Case	030025
CW	H	D	030026
CW	H	Case	030027
CW	D	Case	030028
CCW	A	B	030029
CCW	E	F	030030
CCW	H	J	030031

3.10 Environmental testing. The assembly shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The assembly shall also comply with applicable paragraphs in this section before and after each environmental test. Testing shall be as specified in 4.4.5. Tests shall be conducted in the order specified.

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3.10.1 Vibration.

3.10.1.1 Transportation vibration. The assembly shall be vibrated at the frequencies, gravity unit (g) accelerations, temperatures and times specified in Table IV without evidence of damage such as cracks, chips, separations, or loose parts. The assembly shall comply with the requirements of 3.6, 3.8 and 3.9 after being tested as specified in 4.4.5.1.

Table IV

Frequency Range Hertz (Hz)	Double Displacement or Peak g Level ± 5% Tolerance	Cycle Time to Cover Frequency Range (minutes ± 5)	Total Test Time in Minutes at Given Temperature 80 ± 20°F	160 ± 5°F	-65 ± 5°F
10-20-10	2.0 g	3. ± 0.5	15	10	10
20-50-20	0.036 inch	3. ± 0.5	15	10	10
50-500-50	5.0 g	3. ± 0.5	15	10	10

3.10.1.2 Flight vibration. The assembly shall withstand subjection to vibration over a frequency range of 75 to 1000 to 75 Hertz (Hz) at a peak level of 5 g's ± 5 percent and a temperature of 80 ± 20 degrees Fahrenheit (°F). The vibration shall be applied from a low to a high frequency and returned to a low frequency over a period of 3 ± 0.3 minutes in each of three mutually perpendicular axes, one of which is the direction of forward motion (see Figure 1). The resistance of the circuits specified in 3.5, Table II, shall not exceed the specified value while being tested and shall show no evidence of opening during the test. The test shall be conducted as specified in 4.4.5.2.

3.10.2 Acceleration. The assembly shall withstand without malfunction acceleration of 50 ± 5 g's for 5 seconds in the direction of forward motion (see Figure 1) and of 18 ± 1 g for 15 seconds perpendicular to the direction of forward motion. The assembly shall comply with the requirements of 3.6, 3.8 and 3.9 after being tested as specified in 4.4.5.3.

3.10.3 Shock.

3.10.3.1 Transportation shock. The assembly shall withstand shocks of 75 ± 8 g's peak amplitude with a half amplitude duration of 9 ± 3 milliseconds. The shocks shall be applied three times in each of three mutually perpendicular axes. The assembly shall comply with the requirements of 3.6, 3.8 and 3.9 after being tested as specified in 4.4.5.4.

3.10.3.2 Impact shock. The assembly shall be capable of withstanding impact shocks of 150 ± 15 g's peak amplitude with a half amplitude duration of 4 ± 1 millisecond applied once in the axial direction opposite to that of forward motion (see Figure 1). This shock shall be applied once while the

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shaft is in the cw position and once while the shaft is in the ccw position. The circuit resistance of three sets of normally closed contacts in series (in either cw or ccw shaft positions) shall not exceed 0.12 ohms during or after testing. The test shall be conducted as specified in 4.4.5.5.

3.10.4 High temperature. The assembly shall withstand subjection to 160 + 5°F without damage and shall meet the requirements of 3.6 through 3.9 while at 160 + 5°F. The test shall be conducted as specified in 4.4.5.6.

3.10.5 Low temperature. The assembly shall withstand subjection to -80 + 5°F without damage and shall meet the requirements of 3.6 through 3.9 while at -65 + 5°F. The assembly shall be placed in a container suitable for protection from frosting during this test. The test shall be conducted as specified in 4.4.5.7.

3.11 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.12 Test codes (see 3.5). The contractor shall provide data feedback on punch cards in conformance with Quality Assurance Instructions QAI-PA-1. The data required to be coded, and the code numbers, are specified in the applicable paragraphs in this section.

3.13 Workmanship.

3.13.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.13.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.13.3 Burrs. Parts shall be free from burrs.

3.13.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.13.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.13.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the

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inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 8 complete assemblies (four groups composed of two units each) (see dwg 8821419) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table V. Tests shall be conducted in order specified in Table V.

Table V

First article inspection and tests

Inspection or Test	Group I	Group II	Group III	Group IV	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)	X	X	X	X		X
2. Switch rotation (see 3.6)	X	X	X	X		X
3. Circuit resistance (see 3.8)	X	X	X	X		X
4. Insulation resistance (see 3.9)	X	X	X	X		X
5. Transportation vibration (see 3.10.1.1)	X	X	X	X	X	
6. Acceleration (see 3.10.2)	X	X	X	X	X	
7. Flight vibration (see 3.10.1.2)	X	X	X	X	X	
8. Transportation shock (see 3.10.3.1)	X	X			X	
9. Impact shock (see 3.10.3.2)	X	X			X	
10. High temperature (see 3.10.4)			X		X	

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Inspection or Test	Group I	Group II	Group III	Group IV	Dest	Non-Dest
11. Low temperature (see 3.10.5)				X		X
12. Current capacity (see 3.7)	X	X				X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table VI and Table VII), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table VI
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table VII
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Switch, rotary: fuze option M14 before potting (see dwg 8821419).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Cable length incorrect	Gage
	102. Improper soldering (see 3.11)	Visual
	103. Tie points incorrect or mislocated	Visual
	104. Total shaft rotation incorrect	Visual
	105. Any component missing, damaged or loose	Visual
	106. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Presence of chips or any other foreign matter unless removable	Visual
	202. Evidence of poor workmanship (see 3.13)	Visual

4.3.2.1.2 Switch, rotary: fuze option, after potting (see dwg 8821419).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:	(Connector Defects)	
101.	Wrong connector	Visual
102.	Deformation of connector which would preclude proper mating	Visual/Manual
103.	Portion of connector broken away (minor chips, scratches, burrs, or gouges which do not affect mating are not objectionable)	Visual
104.	Permanently damaged terminal(s) (gouged, nicked or flattened)	Visual
105.	Any pin missing (as evidenced by empty hole in inserts)	Visual
*106.	Height of pin(s) (pins of the same diameter must all appear to be within 1/32 inch in height relative to the adjacent pins)	Visual/Gage
107.	Any pin bent	Visual
108.	Inadequate potting on back of connector	Visual
109.	Improper location and alignment of the four mounting holes and switch shaft, stop pin	Gage
110.	Any component missing or insecurely assembled	Visual/Manual
111.	Any component cracked	Visual
112.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
201.	Foreign matter on inserts unless removable	Visual
202.	Foreign matter on terminals	Visual
203.	Evidence of poor workmanship (see 3.13)	Visual
204.	Marking mislocated, misleading or unidentifiable	Visual
205.	Serialization improper or missing (see 3.4)	Visual

*In the event pin(s) do not appear to be within 1/32 inch in height relative to the adjacent pins, Gage (dwg 8842517) shall be used to determine the variation from the normal height. Reject any pin that varies by more than 0.032 inch max from the normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface of all measurements.

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4.3.2.1.3 Bag, packing, ammunition, for selector fuze, option (see dwg 8822039).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Wrong size bag	Visual
102.	Bag not properly sealed	Visual
103.	Air in bag not minimized before sealing	Visual
104.	Wrong size tape	Visual
105.	Packing material wet	Visual
106.	Incorrect or insufficient cushioning material	Visual
107.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
201.	Marking improper, illegible, smeared, missing or in wrong location	Visual
202.	Evidence of poor workmanship (see 3.13)	Visual

4.3.2.1.4 Container, packing, ammunition, for domestic (CONUS) shipments of special weapon items, packing and marking (see dwg 8808362).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Packing material contaminated	Visual
102.	Incorrect of insufficient cushioning material	Visual
103.	Container damaged	Visual
104.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
201.	Marking improper, illegible, smeared, missing or in wrong position	Visual
202.	Evidence of poor workmanship (see 3.13)	Visual

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4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Switch rotation	Major
b. Continuous current	Major
c. Circuit resistance	Major
d. Insulation resistance	Major
e. Transportation vibration	Major
f. Transportation shock	Major
g. High temperature	Major

Of these tests, e, f and g are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Switch rotation, continuous current, circuit resistance and insulation resistance. Each switch assembly in the lot shall be tested in accordance with 4.4.1 through 4.4.4. If any item fails to comply with 3.6 through 3.9, the item shall be rejected and removed from the lot.

4.3.3.2 Transportation vibration, transportation shock and high temperature. A sample shall be selected at random in accordance with Table VIII. The sample shall be tested in accordance with 4.4.5.1, 4.4.5.4 and 4.4.5.6. If any item fails to comply with 3.10.1.1, 3.10.3.1 and 3.10.4, the lot shall be rejected.

Table VIII

Sample sizes for destructive testing

Number Required to be Delivered	Sample Size for No. Failures
2-180	2
181-500	3
501-800	5
801-1300	7

Lot Size Required = Number Required to be Delivered + Sample Size.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8842556).

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When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Switch rotation. Mount the assembly on a rigid fixture. Measure the torque required to change positions of the switch in both the cw and ccw directions. Operate the switch slowly several times, observe action for possible "dead positions," i.e., positions on which the switch might stop other than its normal positions.

4.4.2 Current capacity.

4.4.2.1 Continuous current. Using a dc current source and the resistance required to limit the current, cause a current of 1.0 + 0.1 ampere to flow through each circuit described in Table I for a period of 5 minutes. Current shall be monitored continuously during the test with instrumentation capable of detecting current interruptions of 20 microseconds duration.

4.4.2.2 Peak current.

4.4.2.2.1 Peak current (cw). Using a dc current source and the resistance required to limit the current, cause a current of 50 plus 0 minus 5 amperes to flow through each circuit described in Table I for a cw shaft position. The voltage drop across the switch contacts shall be monitored continuously during the test with instrumentation capable of detecting circuit interruptions of 20 microseconds duration.

4.4.2.2.2 Peak current (ccw). Using a dc current source and the resistance required to limit the current, cause a current of 50 plus 0 minus 5 amperes to flow through each circuit described in Table I for ccw shaft position. The voltage drop across the switch contacts shall be monitored continuously during the test with instrumentation capable of detecting circuit interruptions of 20 microseconds duration.

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4.4.3 Circuit resistance. A suitable electrical resistance measuring instrument shall be connected to the pins specified.

4.4.4 Insulation resistance. An insulation resistance tester, capable of supplying the required voltage and with which a resistance in megohms can be obtained, shall be used to measure the resistance. The current shall be limited to a maximum of 5 milliamperes to protect human life.

4.4.5 Environmental testing (see 3.10).

4.4.5.1 Transportation vibration. The assembly shall be rigidly mounted to a fixture, capable of being supported readily in each of three mutually perpendicular axes, with one axis being along the direction of forward motion. Vibrate the item at the frequencies, amplitudes, and for the times defined in Table IV in each of three mutually perpendicular axes. Following this test, the assembly shall be tested as specified in 4.4.1, 4.4.3 and 4.4.4.

4.4.5.2 Flight vibration. The item shall be rigidly mounted to a fixture capable of being supported readily in each of three mutually perpendicular axes, with one axes being along the direction of forward motion. The item shall be subjected to vibration over a frequency range of 75 to 1000 Hz at a peak level of 5 g's ± 5 percent and at a temperature of 80 ± 20°F. The vibration shall be applied from a low to a high frequency and returned to a low frequency over a period of 3 ± 0.3 minutes in each of three perpendicular axes. The continuity of the normally used circuits in Table II, under the operating conditions given, shall be measured continuously during the test with instrumentation capable of detecting circuit openings of 20 microseconds duration. During this test the item shall be tested in accordance with 4.4.3.

4.4.5.3 Acceleration. Mount the assembly on the centrifuge so that the forward direction of the assembly is oriented to extend in the radial direction of the centrifuge. Bring the centrifuge up to the rotational speed required to produce the acceleration of 40 ± 5 g's in the forward direction of the assembly. Subject the assembly to 40 ± 5 g's in the forward direction for a period of 5 seconds. Mount the assembly on the centrifuge, so that the forward direction of the assembly extends in a direction perpendicular to the radial direction of the centrifuge. Bring the centrifuge up to rotational speed required to produce 18 ± 1 g in the direction perpendicular to the forward direction of the assembly. Subject the assembly to 18 ± 1 g in this direction for a period of 15 seconds. Following this test, the assembly shall be tested as specified in 4.4.1, 4.4.3 and 4.4.4.

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4.4.5.4 Transportation shock. The assembly shall be rigidly mounted to a fixture capable of being supported readily in each of three mutually perpendicular major axes. The assembly shall be subjected to shocks of 75 ± 8 g's with a half amplitude duration of 9 ± 3 milliseconds three times in each direction of three mutually perpendicular axes for a total of 18 shocks. Following this test, the assembly shall be tested as specified in 4.4.1, 4.4.3 and 4.4.4.

4.4.5.5 Impact shock. The item shall be rigidly mounted to a fixture capable of supporting the item in the forward axial direction. The item shall be subjected to 150 ± 15 g's having a 4 ± 1 millisecond duration at half amplitude applied once in each switch position in the axial direction opposite to that of forward motion. Monitor the normally closed contacts by causing a current of 0.5 amperes dc minimum to be conducted during the test. The circuits shall be monitored continuously during the test with instrumentation capable of detecting circuit openings of 20 microseconds duration.

4.4.5.6 High temperature. The assembly shall be placed in a test chamber and the temperature of the chamber raised to 160 ± 5°F. The assembly shall remain at this temperature for a minimum of 4 hours. At the conclusion of this period and while at 160 ± 5°F, the assembly shall be tested as specified in 4.4.1 through 4.4.4.

4.4.5.7 Low temperature. The assembly shall be placed in a test chamber and the temperature of the chamber lowered to -80 ± 5°F. The assembly shall remain at this temperature for a minimum of 4 hours. The chamber temperature shall then be raised to -65 ± 5°F and maintained for an additional 1 hour period or until stabilization (see 6.4) is reached, whichever is longer. At the conclusion of this period and while at -65 ± 5°F the assembly shall be tested as specified in 4.4.1 through 4.4.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The assembly shall be packaged in accordance with dwg 8822039 and 8808362 using the specified moisture barrier.

5.1.2 Level C. The assembly shall be packaged in accordance with dwg 8822039 and 8808362 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The assembly shall be packed in accordance with dwg 8822039 and 8808362.

5.3 Marking. Marking shall be as specified in accordance with dwg 8822039.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 3.5). Data specified in 3.8, 3.9 and 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1674A dated 24 January 1967.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A137

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